23. Entanglement certification with explicit inconclusive outcomes

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Two quantum systems can be related in ways that no mixture of independent local states can reproduce. That is entanglement. Given a mathematical description of a joint state, this problem asks for evidence that it is entangled, evidence that it is separable, or an honest statement that the available tests cannot decide. It also asks how to quantify one particular aspect of entanglement without mixing different measures together.

Key terms

- **Bipartite system** — A quantum system divided into two identified parts, such as two qubits held by different parties.
- **Density matrix** ρ — An operator describing a quantum state, including statistical mixtures.
- **Separable state** — A mixture of product states of the two parts; correlations in such a decomposition can arise from shared classical information.
- **Entangled state** — A state that has no separable decomposition across the chosen division.
- **Entanglement witness** — An observable nonnegative on every separable state but negative on the state being tested.
- **Partial transpose** — A matrix operation transposing the indices of just one subsystem, useful for testing entanglement.
- **PPT** — Positive partial transpose: a necessary separability condition that is sufficient only in certain small dimensions.
- **Negativity** — A quantity measuring the negative part of the partial-transpose spectrum; zero does not generally imply separability.
- **DPS hierarchy** — Tests based on whether the state can be extended to larger symmetric states; failure can certify entanglement.
- **Concurrence** — A particular entanglement measure with a useful explicit formula for two-qubit states.

Why this matters

Entanglement is an important resource in many quantum-information protocols, but claiming it from an inconclusive test can invalidate a proposed demonstration. Conversely, explicit witnesses show exactly what establishes its presence. Quantitative bounds are useful because different states may possess very different strengths or kinds of quantum correlations.

What progress would mean

The deliverable would make its verdict inspectable: a witness, a decomposition, or unresolved status. That would provide a reliable component for analyzing quantum states, rather than promise a universally easy entanglement test.

Core question

For a finite bipartite density matrix, can one produce a checked entanglement witness or a separable decomposition, and bound a specifically chosen entanglement measure?

Scope and assumptions

Input subsystem dimensions and an exact rational/algebraic density matrix, or a rigorously described uncertainty set. Begin with 2×2 systems; then include 3×3 PPT-entangled examples. Validate Hermiticity, positivity and unit trace before classification.

The primary decision outputs are entangled, separable, or unresolved. Restrict concurrence and its entanglement-of-formation formula to two qubits. Treat general distillable entanglement and multipartite classification as separate research problems.

Mathematical target

Compute partial transpose in a fixed tensor-index convention. Negativity is $\mathcal{N}(\rho) = (\|\rho^{T_B}\|_1 - 1)/2$. For entanglement give W satisfying $\langle a \otimes b|W|a \otimes b\rangle \geq 0$ for all product vectors and $\mathrm{Tr}(W\rho)<0$. For separability give $\rho = \sum_j p_j |a_j b_j\rangle\langle a_j b_j|$ with p_j≥0 and Σp_j=1, or another sufficient theorem in its valid dimensions.

PPT/DPS feasibility is generally only a necessary separability test. DPS infeasibility with a validated dual can certify entanglement; passing a finite level generally remains unresolved.

Required outputs and proof obligations

Export input validation, partial-transpose spectrum enclosures, measure intervals, and a witness/decomposition with an independent checker. Normalize witnesses before optimizing them. For an uncertain input, require the detection margin to hold throughout the uncertainty set.

Validation and rejection controls

Use Bell states, product states, and $\rho_p = p|\psi^-\rangle\langle\psi^-| + (1-p)I/4$, $0 \leq p \leq 1$. In this declared convention separability holds for $p \leq 1/3$. Verify negativity 1/2 and concurrence 1 for a Bell state. Include a 3×3 PPT-entangled control so zero negativity is not mislabeled separability.

Milestones and research extension

Baseline: checked two-qubit decisions and measure values. Strong: a validated DPS witness beyond PPT and a separable-decomposition example. Research extension: robust uncertainty-set certification or normalized optimal witnesses on a stated family.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Doherty, Parrilo and Spedalieri, A complete family of separability criteria — Symmetric-extension hierarchy and dual entanglement witnesses.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\mathcal{N}(\rho) = \frac{\|\rho^{T_B}\|_1 - 1}{2}, \qquad \langle a \otimes b|W|a \otimes b\rangle \geq 0, \quad \mathrm{Tr}(W\rho) < 0.$$

$$\rho = \sum_j p_j |a_j b_j\rangle\langle a_j b_j|, \qquad p_j \geq 0, \quad \sum_j p_j = 1.$$

The witness conditions certify entanglement; the normalized product-state decomposition certifies separability. In general, passing finite PPT/DPS tests remains unresolved. For the two-qubit family $\rho_p = p|\psi^-\rangle\langle\psi^-| + (1-p)I/4$, $0 \leq p \leq 1$, separability holds exactly for $p \leq 1/3$.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — relaxation direction (36–40).** Passing PPT or a finite DPS level does not generally prove separability. The replacement requires a sufficient certificate or reports unresolved.
2. **Major — "optimal witness" unspecified.** Multiplying a witness by a positive constant scales its objective without changing detection. An optimization needs a normalization and a declared class of witnesses.
3. **Major — measures conflated (23–29, 87–107).** Different entanglement measures do not give a complete interchangeable characterization, and the two-qubit concurrence formula is not a general mixed-state solution.

4. **Major — benchmark convention (817).** A Werner-state threshold such as p=2/3 is meaningless without the mixing definition; for the singlet-weight convention used here the threshold is 1/3. Explicit inputs remove that ambiguity and expose false positives in the checker.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a finite bipartite density matrix, can one produce a checked entanglement witness or a separable decomposition, and bound a specifically chosen entanglement measure?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: checked two-qubit decisions and measure values. Strong: a validated DPS witness beyond PPT and a separable-decomposition example. Research extension: robust uncertainty-set certification or normalized optimal witnesses on a stated family.

Sources supporting the corrections

- Doherty, Parrilo and Spedalieri, A complete family of separability criteria — Symmetric-extension hierarchy and dual entanglement witnesses.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.

- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.